

# Moamen Moustafa Basyoni

Senior C# / .NET Core Developer | Distributed Backend Systems

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## Professional Summary

Software Engineer with around three years building and modernizing backend services on modern .NET (.NET 8/9/10, ASP.NET Core, Minimal APIs) — extending and refactoring large, mature codebases and operating distributed, event-driven systems with RabbitMQ, Temporal, and Consul. Strong in REST and gRPC APIs, asynchronous and concurrent programming, Entity Framework Core over SQL Server and PostgreSQL, Docker, and CI/CD. Early adopter of AI tooling and agentic development workflows.

## Professional Experience

### Senior C# / .NET Core Developer — Andalusia Health Business Solution (AHBS)

Dec 2023 – Present

Alexandria, Egypt

- Extended, refactored, and **modernized a large, mature codebase** for the Andalusia hospital group across **Egypt and Saudi Arabia**, migrating healthcare / Hospital Management (HMIS) modules from **.NET Framework 4.5 / ASP.NET Core 2.2** toward modern .NET with **Clean Architecture** across 5 core and 7+ additional modules (Blood Bank, Supply Chain, Bed Management, Hospital Structure).
- Designed and built **REST and gRPC APIs** for distributed, service-based hospital modules, and designed efficient **data access layers** with Entity Framework Core over SQL Server and PostgreSQL.
- Built distributed, **event-driven** communication with **RabbitMQ** and long-running distributed transactions with **Temporal** Saga orchestration, improving 10+ inter-service interaction patterns.
- Implemented **service discovery and dynamic routing** with **Consul** and Consul Template behind an **Nginx API Gateway** with health-aware load balancing and parallel service versioning — cutting deployment downtime from 2 hours to 15 minutes.
- Built a runtime **Business Rule Engine** for injected business logic, improving rule-execution performance by 45% (measured with **BenchmarkDotNet**).
- Used **asynchronous programming (async/await)** throughout for input/output-bound service workloads, and delivered **CI/CD pipelines in Azure DevOps** with multi-repository Git Flow and **Docker** across Dev, Staging, and Pre-Live environments.
- Built internal **Model Context Protocol (MCP) servers** and automated code-review / pull-request analysis tooling, and led an **AI-assisted and agentic development** onboarding program for 25 engineers across 6 teams.

### Backend Engineer — Trastain

Feb 2025 – Aug 2025

Cairo, Egypt (Remote) · Part-time · Real-estate technology (PropTech) marketplace

- Built a **PropTech marketplace backend from scratch**, including shared-property booking, policy enforcement, and preference / culture-fit matching.
- Developed **retrieval-augmented generation (RAG)** property matching with actionable booking recommendations and transaction security for booking execution.
- Automated build and deployment with **GitHub Actions** CI/CD workflows to Azure Web Apps and Azure Static Web Apps across Dev and Staging.

### Software Engineer — MYM (Make Your Miracle)

Jun 2023 – Oct 2023

Giza, Egypt · SaaS / e-commerce / booking

- Developed backend systems for e-commerce and booking platforms, delivering end-to-end from requirements analysis through implementation, deployment, and support.
- Deployed applications to Azure App Services.

## Technical Skills

**Languages:** C#, SQL, JavaScript, TypeScript, Python

**.NET & Backend:** .NET 8/9/10, .NET Framework 4.5, ASP.NET Core, ASP.NET Core MVC, Minimal APIs, Entity Framework Core, .NET Aspire, REST APIs, gRPC, Webhooks

**Architecture:** Clean Architecture, Microservices, Distributed Systems, Service-Oriented Architecture, Saga orchestration, API Gateway, Business Rule Engines

**Distributed Systems, Messaging & Concurrency:** RabbitMQ, Temporal, Consul (service discovery, health checks, KV), Nginx, load balancing, event-driven architecture; async/await, asyncio, parallel service versioning

**Data:** SQL Server, PostgreSQL, Neo4j, Elasticsearch

**Identity & Security:** OpenID Connect (OIDC), OAuth2, OpenIddict, JSON Web Tokens (JWT), scope- and claim-based authorization

**DevOps & Version Control:** Azure DevOps (Cloud & On-Premise), GitHub Actions, CI/CD pipelines, Docker, Git, Git Flow, multi-repository CI/CD, TFS / TFVC, Azure Web Apps, Azure Static Web Apps

**AI-Assisted & Agentic Development:** Claude Code, GitHub Copilot, Codex, Model Context Protocol (MCP), LangGraph, CrewAI, RAG, GraphRAG

**Testing & Performance:** pytest, pytest-asyncio, automated testing, BenchmarkDotNet, performance tuning

## Education

**B.Sc. Computer Engineering — Behera Higher Institute (BHI), Alexandria, Egypt**

Grade: B+ · GPA 3.2 / 4.0

2023

## Publication & Languages

**IoT-Enabled E-Prescription Management and Dispensing Machine Monitoring** — IEEE, 2024.

Arabic (Native) · English (B2, Upper Intermediate)